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mCNN-GenEfflux: enhanced predicting Efflux protein and their super families by using generative proteins combined with multiple windows convolution neural networks

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ABSTRACT

Efflux transporters play a critical role in bacterial antibiotic resistance by facilitating the removal of harmful substances. These are classified into five distinct families: ABC, MFS, MATE, RND, and SMR. The significant sequence variability among these families, coupled with insufficient functional annotation, presents considerable challenges for traditional categorization methods. We hypothesize that integrating of ProtGPT2-generated efflux protein sequences with a multi-window convolutional neural network (mCNN) is proposed to enhance classification accuracy by effectively capturing local motifs and broader evolutionary patterns often overlooked by previous methods. Generative models like ProtGPT2 are effective for this purpose, as they generate a variety of sequence variants that reflect patterns observed in natural efflux families, thereby minimizing issues related to data scarcity. The proposed GenEfflux framework, unlike alignment-based methods such as HHblits and single-feature CNNs, combines generative sequence expanding with multi-scale evolutionary feature extraction via Position-Specific Scoring Matrices (PSSMs), thereby enhancing the understanding of sequence-function relationships. In comparative evaluations, GenEfflux consistently outperformed the baseline deepEfflux model across all Efflux transporter classes. In Class B, sensitivity increased from 0.5385 to 0.9999, and the Matthews correlation coefficient (MCC) rose from 0.4397 to 0.9327. In Class C, accuracy improved from 0.8977 to 0.9668, alongside an increase in MCC from 0.7668 to 0.9331. The findings demonstrate that sequences generated by ProtGPT2 possess functional relevance and improve classification effectiveness. GenEfflux suggestions a comprehensive framework for enhancing efflux protein analysis and advancing research on antibiotic resistance.

1. Introduction

Recent advances in artificial intelligence (AI) have transformed major scientific fields, including natural language processing (NLP), computer vision, and bioinformatics (Vaswani et al., 2017; Brown et al., 2020). Among these, generative models have emerged as powerful tools capable of generating realistic and meaningful data such as human-like text, images, and biological sequences by learning patterns from large datasets (Raffel et al., 2020a; Devlin et al., 2019; Radford, 2018). Based on the Transformer architecture, these models excel in generating logical and consistent outputs such as human-like text, images, and biological sequences (Devlin et al., 2019; $author1$ et al., 2023; Raffel et al., 2020b). Initially developed for linguistic tasks like text generation, summarization, and translation, models such as GPT (Radford, 2018), LLaMA ($author1$ et al., 2023), T5 (Raffel et al., 2020b), and BERT (Devlin et al., 2019) have set benchmarks in NLP by enabling systems to generate and comprehend human-like text (Yenduri et al., 2024). Recent adaptations of generative models in bioinformatics focus on the design and analysis of biological sequences (Simon et al., 2024). In protein engineering, a vital domain of biotechnology and medicine, generative models tackle the limitations of sparse experimental data by generating novel protein sequences that exhibit improved functionality and diversity (Devlin et al., 2019; Mardikoraem et al., 2023; Madani et al., 2020a). Generative methods demonstrate significant potential for applications in drug discovery, enzyme engineering, and synthetic biology. For instance, they can improve protein diversity by generating

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sequences that exhibit enhanced drug-binding affinity or catalytic efficiency (Wu et al., 2021; Alexander et al., 2021; Johnston et al., 2023). ProtGPT2 serves as a generative model designed explicitly for protein sequences (Ferruz and Schmidt, 2022). In contrast to conventional approaches that depend on curated datasets, ProtGPT2 utilizes comprehensive unsupervised training to generate physiologically realistic protein sequences. The sequences demonstrate genuine amino acid patterns and comply with the principles of protein structure (Ferruz, 2022). Optimizing the synthesis of customized sequences through the precise adjustment of ProtGPT2 for particular protein families enhances biochemical properties and functional diversity (Madani et al., 2020a). These advancements address critical gaps in protein engineering, facilitating accurate protein design and functional assessment. These developments offer numerous opportunities across industry, research, and healthcare (Simon et al., 2024; Ferruz and Schmidt, 2022; Wu et al., 2024). Efflux proteins play a critical role in bacterial resistance by actively transporting harmful substances, such as antibiotics, out of bacterial cells (Ahmad et al., 2024). Efflux pumps, located in bacterial membranes, confer protection to microorganisms against antimicrobials by lowering intracellular drug concentrations. This mechanism plays a role in multidrug resistance, biofilm formation, quorum sensing, and the detection of antibiotic resistance (Huang et al., 2022; Hajiagha and Kafil, 2023; Lewis et al., 2023). These pumps extrude a diverse array of substances, such as heavy metals, pollutants, plant compounds, quorum-sensing signals, and bacterial metabolites (Blanco et al., 2016; Chitsaz and Brown, 2017; Chughtai et al., 2023). Efflux proteins are classified into five primary groups according to their structural and functional properties, as depicted in Fig. 1 (Wellem Taju et al., 2018). The ABC family utilizes ATP to translocate substrates (Barreto and Zoghbi, 2024; Thomas et al., 2020; Fatht and Kolter, 1993), MFS proteins transport ions and drugs (Pao et al., 1998), MATE pumps expel xenobiotics using Na^+ or H^+ gradients (Kusakizako et al., 2020), and SMR and RND families mediate multidrug efflux, contributing to resistance (Kermani et al., 2020; Kavanaugh et al., 2024). Efflux proteins from SMR, MFS, RND, and MATE families are highly diverse, poorly conserved, and often underrepresented in protein databases, significantly limiting the effectiveness of traditional alignment-based or discriminative approaches due to their inherent sequence variability (Wellem Taju et al., 2018; Kusakizako et al., 2020; Li et al., 2015; Pearson, 2013). SMR proteins show significant sequence variability and have been subdivided into functional subclasses (Bay et al., 2008), while MFS pumps like KpsrMFS exhibit atypical regulation and influence reactive oxygen species and growth (He et al., 2024). MATE transporters, which remove a wide range of xenobiotics using Na^+ or H^+ gradients, also exhibit structural and substrate diversity (Radchenko et al., 2015). The RND superfamily spans multiple taxa and comprises mechanistically distinct subclasses, underscoring the need for flexible and generalizable modeling strategies (Sharma et al., 2023; Tseng et al., 1999). Generative models like ProtGPT2 address these challenges by learning directly from large-scale protein sequence data and generating novel, biologically plausible sequences without relying on sequence alignment (Madani et al., 2020a). Recent work has shown that retrained general-purpose large language models (LLMs), such as LLaMA ($author1$ et al., 2023) and Mistral, can rival established domain-specific protein models including ProGen (Madani et al., 2020b), and ProtGPT2 even when trained on relatively small protein datasets (Zeinalipour et al., 2024). Their ability to populate unexplored regions of protein space makes them particularly powerful for modeling complex and low-homology protein families such as efflux pumps (Nijkamp et al., 2023). Protein structure prediction remains a core challenge in computational biology, vital for understanding biological function and guiding therapeutic development. Determining a protein's three-dimensional conformation from its amino acid sequence is difficult due to the vast diversity of structural folds and the complex, non-linear interactions involved in protein folding (Jumper et al., 2021; Dill and MacCallum, 2012; Katoh and Standley, 2013). Structure prediction methods are typically classified into two categories: alignment-based and alignment-free techniques. Alignment-based methods rely on identifying conserved regions through sequence alignments. While often accurate, they are computationally intensive and can be sensitive to sequence rearrangements or transpositions, limiting their utility for highly variable or novel sequences (Zielezinski et al., 2017). In contrast, alignment-free methods bypass sequence comparison by applying strategies like K-mer segmentation and machine learning. These approaches offer improved speed and scalability but may sacrifice important sequence-order information and often require dataset-specific parameter tuning (Li et al., 2018). For example, deep learning frameworks such as DeepFam represent sequences using binary matrices, but these representations may fail to capture the full complexity of biochemical and structural properties (Shahid Malik and Ou, 2023). Recent innovations have applied natural language processing (NLP) techniques to protein sequences by treating them as structured "languages" (Malik et al., 2024; Rives et al., 2021). This conceptual shift enables richer contextual modeling and has improved performance in several predictive tasks. However, the effectiveness of NLP-based models is still constrained by the limited availability of comprehensive, high-quality protein datasets (Torrisi et al., 2019). To incorporate evolutionary context, position-specific scoring matrices (PSSMs) are

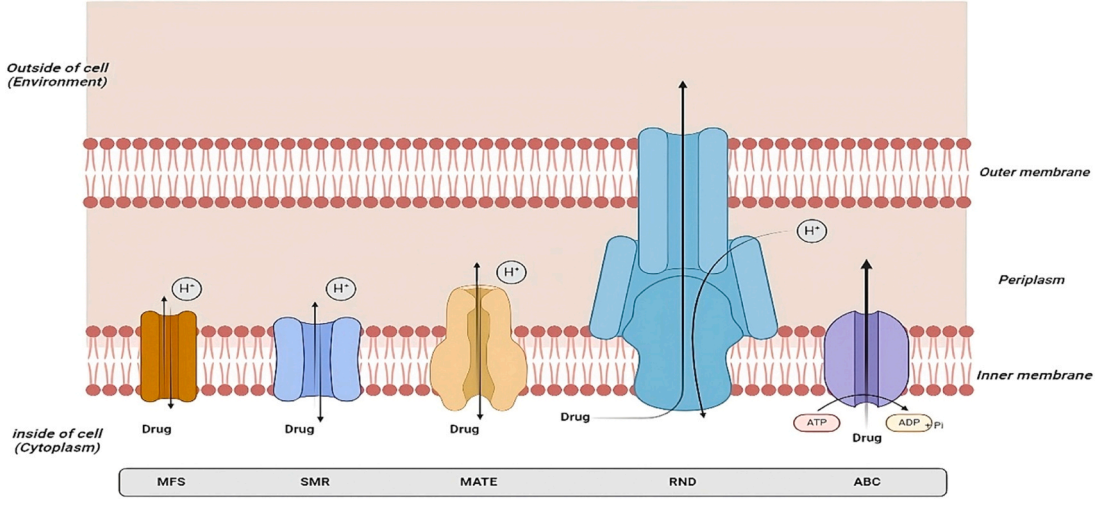


Fig. 1. Efflux proteins are transporters found in the cytoplasmic membrane of various cell types. These proteins play an essential role in removing substances from the cell and belong to five families of efflux proteins. Each family uses different types of chemical energy to accomplish its specific activities.

frequently used in structure prediction. When paired with convolutional neural networks (CNNs), PSSMs can enhance predictive accuracy by highlighting conserved residues. Nevertheless, these methods may lose sequence-order fidelity during aggregation and struggle to distinguish sequences with similar amino acid distributions but different functions (Yang et al., 2018; Milton et al., 2016). Alignment-based sequence search methods, including HMM approaches like HHblits, frequently fail to identify hidden or distant homology among highly divergent protein families (Gao and Skolnick, 2020b; Llinares-López et al., 2023). Deep learning models that use single-window convolutional architectures may fail to capture hierarchical or distributed sequence patterns due to the limitations of their fixed receptive field dimension (Gligorijević et al., 2021). Traditional classifiers dependent on predefined sequence motifs or customized characteristics are usually unable to generalize to novel or low-homology proteins, as these manually designed features often oversimplify complicated sequence–function relationships (Nguyen and Hy, 2025).

In this study, we assess how closely sequences generated by ProtGPT2 resemble natural efflux proteins in terms of sequence similarity and biological characteristics. By combining ProtGPT2 with a multi-window CNN (mCNN) architecture and PSSM-based feature extraction, we improve the model's ability to detect functional patterns in low-similarity sequences. PSSMs provide evolutionary context, while the mCNN model applies multiple filter sizes to capture both local and global sequence signals. This approach enhances feature representation and classification accuracy. GenEfflux introduces several key innovations over DeepEfflux, including the use of ProtGPT2-generated sequences, multi-window feature extraction, and interpretability analysis via UMAP and Grad-CAM, along with improved handling of imbalanced datasets. These features collectively enhance model interpretability, generalization, and computational efficiency.

2. Methods and materials

Fig. 2 highlights the comprehensive structure of our methodology, consisting of five primary components: data collection, sequence synthesis, feature extraction, multi-window analysis, and classification. The method begins with protein sequences sourced from the Transporter Classification Database (TCDB). These sequences are clustered into Classes A, B, and C using **PSI-BLAST**, which identifies homologous groups through iterative searches based on evolutionary similarity. This clustering establishes the foundation for supervised learning and model evaluation. Next, ProtGPT2 is fine-tuned on efflux transporter families, enabling the generation of novel sequences that reflect the characteristics of the five principal efflux transporter superfamilies. Position-specific scoring matrices (PSSMs) are then used to evaluate the generated sequences, helping to detect conserved motifs and assess structural integrity. These PSSM profiles are subsequently analyzed using a multi-window convolutional neural network (CNN), which captures both local and global sequence patterns. Finally, a fully connected neural network (FCNN) classifies the sequences according to their functional and structural attributes. This pipeline demonstrates ProtGPT2's capacity to generate physiologically relevant sequences, setting a benchmark for protein synthesis facilitated by artificial intelligence.

2.1. Data collection

We obtained protein sequences by concentrating on the five primary efflux transporter superfamilies: ABC, MATE, MFS, RND, and SMR unique structural and functional characteristics of these families facilitate the creation of a specialized and high-quality dataset from to extract sequences specific to each efflux superfamily, we used the extractTCDB.pl script provided by the TCDB tools suite TCDB (Altschul et al., 1997). This strategic approach enhanced model performance and facilitated a comprehensive comparative analysis.

2.2. Experimental setup for fine-tuning ProtGPT2 on PRotein Sequences

The ProtGPT2 model was fine-tuned separately for each efflux family to generate efflux protein sequences, using a high-quality dataset that included sequences from multiple efflux protein families, formatted in FASTA. The dataset was randomly permuted, with 80 % designated for training and 20 % for validation. The nferruz/ProtGPT2 model and its corresponding tokenizer were utilized in a pre-trained condition. The tokenizer was set to pad sequences with the end-of-sequence token and to truncate sequences to a maximum length of 1400 characters. Tokenized sequences were transformed into Hugging Face Dataset objects and readied for training using PyTorch. Dynamic padding was implemented utilizing Hugging Face's data collator for language modeling, with causal language modeling activated. The training and validation processes employed PyTorch DataLoader objects, starting with a batch size of 1 for debugging purposes. A batch size of two per device was utilized for both training and evaluation. Gradient accumulation was employed, with gradient_accumulation_steps set to 8, to attain a larger effective batch size. The model was trained for 10 epochs with a learning rate of 1e-6. Checkpoints were documented every 200 steps, with a maximum of five checkpoints retained. Model evaluation was conducted at intervals of 200 steps, whereas training records were recorded every 100 steps. Four worker threads were implemented to improve data loading efficiency.

2.3. Efflux protein sequence generation using ProtGPT2

We improved ProtGPT2 for each efflux superfamily by refining the training dataset to eliminate redundant sequences. A threshold of 30 % pairwise sequence identity was established using **PSI-BLAST** to ensure the inclusion of only distinct and diverse sequences within each family derived from the TCDB. This standard is commonly used to reduce replication and is founded on the traditional notion that protein pairings with less than 30 % identity exist in the "twilight zone," where structural or functional similarities are unclear (Rost, 1999). A perplexity threshold of 90 was empirically selected to filter out sequences with high complexity and low generation confidence during the sequence generation process. Only sequences with perplexity lower than 90 were retained, as these are more likely to resemble natural proteins with biologically relevant features. This cutoff follows empirical guidelines from the ProtGPT2 developers, who recommend classifying generated sequences by their perplexity and favoring those with lower values to enhance plausibility. Although a definitive perplexity threshold has not yet been established, lower perplexity scores generally correspond to sequences exhibiting a more "natural" character and correlate with higher AlphaFold2 pLDDT and pTM confidence scores. Structural validation using AlphaFold in this study confirmed that sequences below the threshold showed significantly improved folding confidence (see Supplementary File Table S4). This approach aligns with previous studies supporting the use of perplexity as an effective filter for high-quality generated sequences (Ferruz and Schmidt, 2022). The dataset was obtained from the Transporter Classification Database (TCDB) and included 992 MFS, 18 SMR, 102 MATE, 248 RND, and 2394 ABC efflux protein sequences. After removing redundancy using PSI-BLAST with a 30 % pairwise identity threshold, the non-redundant sequences were further refined, resulting in 533 (MFS), 8 (SMR), 64 (MATE), 94 (RND), and 897 (ABC) sequences. The dataset was partitioned, allocating 80 % for model fine-tuning and 20 % for independent testing. The training set undergoes additional validation through 5-fold cross-validation (CV) during model training to ensure robustness and reduce overfitting. The dataset had imbalanced efflux families, with SMR (18 sequences) much smaller than ABC (2394 sequences. During fine-tuning, the model was trained on individual efflux families, in the fine-tuning process, 426 sequences from MFS, 6 from SMR, 41 from MATE, 75 from RND, and 717 from ABC were used. ProtGPT2 generated 630, 107, 142, 256, and 1117 synthetic sequences for the MFS, SMR, MATE, RND, and ABC

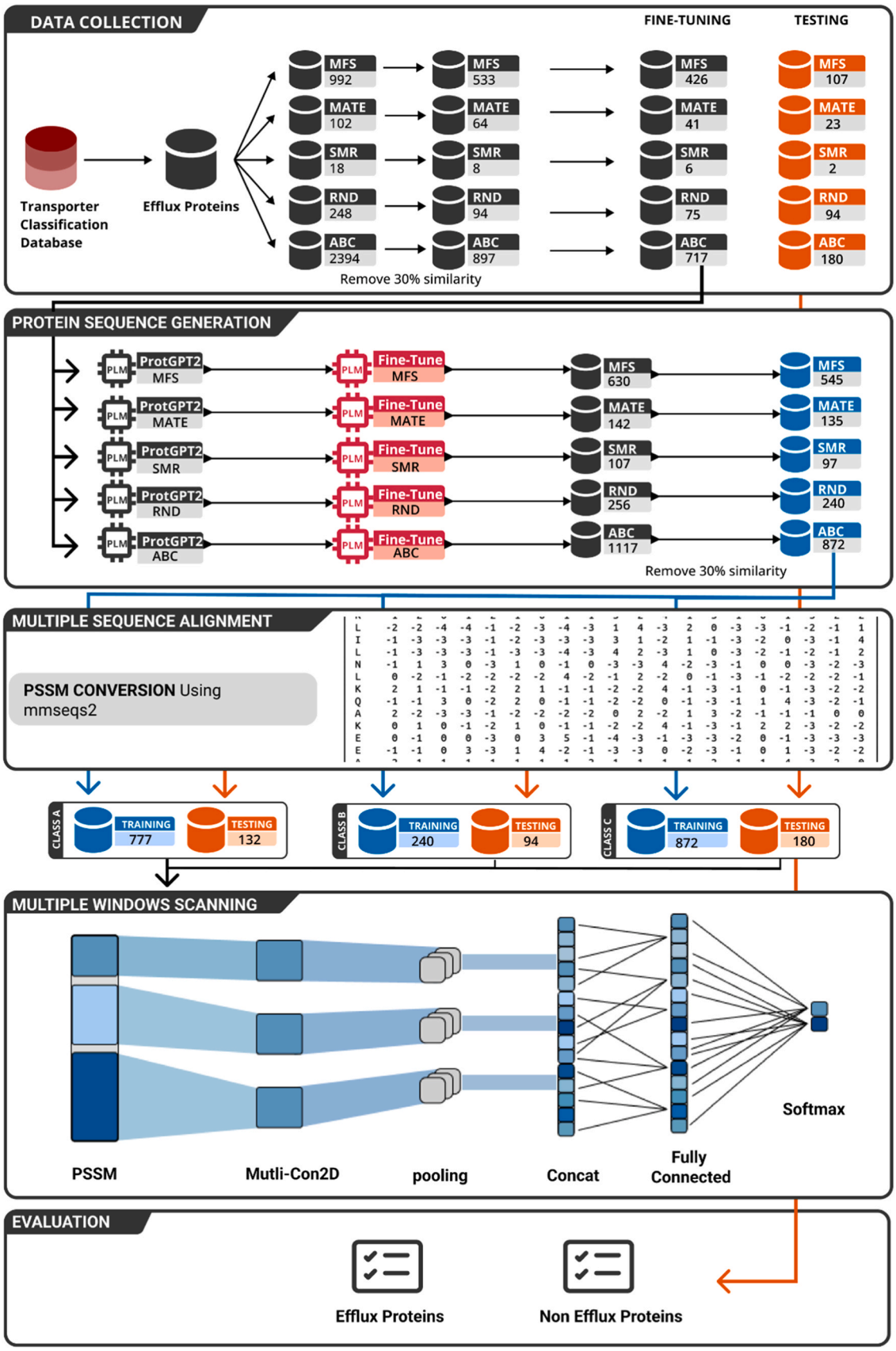


Fig. 2. The efflux protein model combines a protein generation framework with PSSM profiles and a multi-. window scanning approach. This integration allows for effective feature extraction and pattern recognition,. ensuring precise classification of efflux proteins.

families, respectively. Among the analyzed sequences, 545, 97, 135, 240, and 872 exhibited less than 30 % identity, thereby meeting the established criterion. For classification the sequences were classified into three functional categories: Class A (MFS, SMR, MATE) with 777 training and 102 test sequences; Class B (RND) with 240 training and 19 test sequences; and Class C (ABC) with 872 training and 95 test sequences. No class weighting or data augmentation was applied during training to evaluate the model's baseline performance under natural class distributions. Despite the imbalance, the model demonstrated consistent performance, and future work will investigate imbalance mitigation techniques such as oversampling and class weighting.

Multiple sequence alignment

Multiple Sequence Alignment (MSA) is a method used to align three or more biological sequences, such as proteins or nucleotides, to discern areas of similarity and divergence among the sequences. Alignments in bioinformatics are essential for the construction of phylogenetic trees, the prediction of protein function and structure, and the elucidation of evolutionary relationships. Identifying homologous residues or nucleotides assists researchers in pinpointing conserved regions that are frequently functionally or structurally significant. The Position-Specific Scoring Matrix (PSSM) clarifies the evolutionary information present in protein sequences (Nguyen et al., 2022). Multiple sequence alignment (MSA) aligns protein sequences, facilitating the identification of conserved and divergent regions. The alignment technique generates a consensus sequence, which is then employed to determine amino acid frequencies at each alignment position (Ou et al., 2023; Steinegger, 2017). The frequencies are structured into a position-specific scoring matrix (PSSM), represented as a N × 20 matrix for each query protein. N represents the sequence length, with each position containing 20 scores indicative of amino acid conservation. The PSSM score measures the probability of an amino acid's presence at a given position, with elevated scores indicating greater conservation. This study employed the MMseqs2 (Gao and Skolnick, 2020a) tool to generate PSSM profiles from the dataset. PSSM scores, indicated by positive or negative values, define sequence patterns in biological data. A negative value indicates infrequent substitutions, while positive values signify more frequent alterations at specific sites within the protein family.

2.4. Model architecture: multiple-windows convolutional neural network

We developed a multi-window two-dimensional convolutional neural network (2D-CNN) specifically designed to extract biological features from protein sequences using Position-Specific Scoring Matrix (PSSM) inputs shown in Fig. 3. Each Position-Specific Scoring Matrix (PSSM) is represented as a matrix with dimensions N × 20, where N denotes the length of the protein sequence and 20 corresponds to the standard amino acids. These matrices incorporate evolutionary conservation information and serve as informative 2D representations suitable for convolution-based analysis. Our model employs six parallel convolutional branches, each with a distinct kernel size: 1 ×w, where w ∈ {2,4,8,12,16,20}. These convolutional filters slide horizontally across the amino acid feature space, enabling the extraction of both short-range motifs and long-range dependencies. All convolutional layers utilize a stride of (1, 1) and implement 'valid' padding, resulting in a corresponding decrease in output dimensions without the use of zero-padding. Following convolution, each feature map undergoes processing by a Global Max Pooling layer, which combines the feature map into a single scalar by keeping the most significant activation. The outputs from all window sizes are flattened and concatenated into a single feature vector, thereby integrating information from various receptive fields. A dropout layer with a rate of 0.7 is utilized to reduce overfitting. A fully connected dense layer is implemented, comprising 512 neurons activated by ReLU. A softmax output layer with two units, representing binary class labels, is employed for the final classification. This architecture allows the model to integrate evolutionary data from PSSM profiles with multi-scale convolutional insights, thereby enhancing its effectiveness in assessing ProtGPT2-generated sequences in comparison to native efflux proteins.

3. Performance metrics

This study examined the efficacy of a proposed multiple-windows convolutional neural network (CNN) by utilizing position-specific scoring matrix (PSMS) profiles as input. A 5-fold cross-validation approach was employed with the training data to enhance generalizability and robustness. This procedure provided a comprehensive evaluation of model performance prior to testing on an independent set. The model's efflux protein classification accuracy was evaluated using four primary criteria: True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN). TN accurately classified non-efflux proteins, while TP correctly identified efflux proteins. Conversely, FN indicated instances of absent efflux protein classifications, whereas FP referred to cases where non-efflux proteins were incorrectly classified as efflux proteins. The percentage of correctly identified efflux proteins is measured, with sensitivity being a critical statistic. Specificity is assessed through the accuracy of non-efflux protein identification. Precision, as defined in Formula 3, quantifies the ratio of correctly predicted efflux proteins to the total number of positive predictions, thereby illustrating the reliability of positive classifications. The classification accuracy of Formula 4 indicates the ratio of correct predictions to the total predictions made. The evaluation of model performance was conducted through the use of the Matthews correlation coefficient (MCC) and the area under the curve (AUC). In the context of imbalanced datasets, AUC quantifies the model's classification performance across different thresholds, whereas MCC provides a balanced evaluation of predictive accuracy (Formula 5). This assessment demonstrates the model's accuracy in detecting efflux protein interactions, thereby improving our understanding of its precision and potential applications in efflux-related predictions.

$$Sensitivity = \frac{TP}{TP + FN} \tag{1}$$

$$Specificity = \frac{TN}{TN + FP} \tag{2}$$

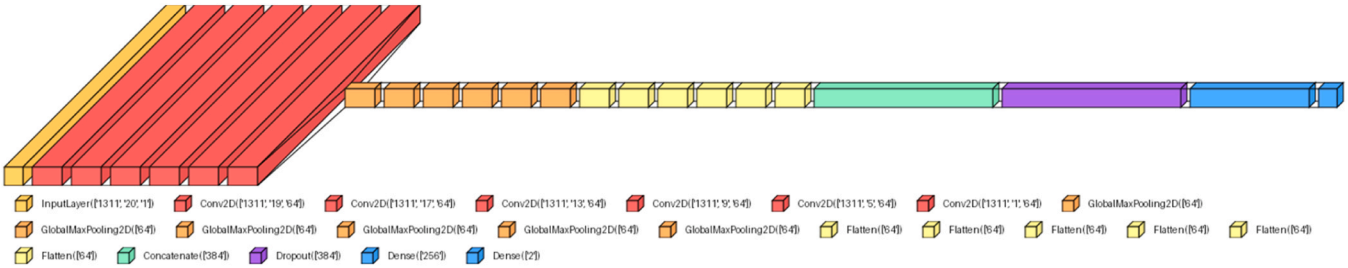


Fig. 3. Multi-kernel CNN architecture for Efflux sequence classification.

$$Precision = \frac{TP}{TP + FP} \tag{3}$$

$$Accuracy = \frac{TP + TN}{(TP + FP + TN + FN)} \tag{4}$$

$$MCC = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \tag{5}$$

4. Results and discussion

We validated the proposed Position-Specific Scoring Matrix (PSSM) on the training dataset using a five-fold cross-validation approach using a multi-window convolutional neural network (CNN) architecture. This methodology enhanced the model during training, resulting in superior performance on the final assessment dataset. The model exhibited improved performance across all efflux protein families, as confirmed by five-fold cross-validation and independent testing. Table 1 displays the performance metrics for each class during the cross-validation process. The model for Class A demonstrated a sensitivity of 0.9151, a specificity of 0.9159, and an accuracy of 0.9156. The Matthews Correlation Coefficient (MCC) was 0.8294, and the Area Under the Curve (AUC) was 0.9623. In Class B, sensitivity rose to 0.9660, specificity to 0.9640, and accuracy to 0.9664, resulting in an MCC of 0.9330 and an AUC of 0.9867. Class C exhibited a sensitivity of 0.9288, a specificity of 0.9406, an accuracy of 0.9346, a Matthews correlation coefficient (MCC) of 0.8693, and an area under the curve (AUC) of 0.9773. Table 1 displays the results of an independent test performed on the efflux proteins. In this evaluation, Class A exhibits a sensitivity of 0.9672, indicating its robust ability to identify the efflux proteins. The specificity for Class A was 0.9487. This indicates that the model successfully identifies the most positive instances, achieving a high accuracy rate of 0.9568. The MCC for this class was 0.9131, while the AUC was 0.9737. Class B exhibited a sensitivity rate of 0.9474 and an effective specificity of 0.9231. The accuracy was measured at 0.9999, with a Matthews correlation coefficient (MCC) of 0.9327 and an area under the curve (AUC) of 0.9636. The results demonstrate effective performance; however, the correlation is marginally lower than that of Class A. Class C exhibited robust performance, with a sensitivity of 0.9667 and a specificity of 0.9621. Class C attained an accuracy of 0.9642, a Matthews Correlation Coefficient (MCC) of 0.9280, and an Area Under the Curve (AUC) of 0.9812. The results indicate the model's strong ability to distinguish between efflux protein classes utilizing PSSM features. The generated sequences display conserved motifs and patterns similar to those observed in natural efflux proteins; however, structural modeling and experimental validation have yet to be conducted. Consequently, claims regarding structural or functional similarity are unsure and require further investigation. Generated sequences were compared to known efflux proteins using BLASTp against the UniProt database. Sequences with $>$ 50 % identity and E-values were considered significantly similar. The top matches are listed in Supplementary Table S4.

4.1. Performance evaluation of single- and multi-window CNN architectures with filter depth variation

The impact of architectural variations on GenEfflux performance was assessed by examining three primary configurations across Classes A, B, and C: (i) single-window convolution with a window size of 20, (ii) multi-window convolution utilizing various kernel sizes (Brown et al., 2020; Devlin et al., 2019; Yenduri et al., 2024; Wu et al., 2021; Ferruz, 2022; Hajiagha and Kafil, 2023), and (iii) modifications in the number of convolutional filters. The evaluation of performance was conducted using five standard classification metrics: sensitivity, specificity, accuracy, Matthews Correlation Coefficient (MCC), and area under the ROC curve (AUC), as detailed in Table 2. The multi-window CNN (mCNN) architecture markedly improved classification performance for Class A (MFS, SMR, MATE) in comparison to the single-window CNN baseline. The mCNN demonstrated a sensitivity of 0.9672, an accuracy of 0.9532, and a Matthews Correlation Coefficient (MCC) of 0.9061. Enhancements were achieved by increasing the filter count to 128, thereby improving feature extraction and yielding a peak accuracy of 0.9568, a Matthews correlation coefficient (MCC) of 0.9131, and an Area Under the Curve (AUC) of 0.9888. These findings demonstrate the efficacy of employing multi-scale convolutional filters alongside more intricate architectures to discern distinct sequence patterns. In Class B (RND), all model configurations demonstrated a sensitivity of 0.9999. The single-window CNN model, utilizing 64 convolutional filters, demonstrated superior specificity compared to the multi-window CNN (mCNN), attaining a score of 0.9615. The model achieved an accuracy of 0.9718, a Matthews correlation coefficient (MCC) of 0.9327, and an area under the curve (AUC) of 0.9727 in this domain. The findings suggest that, for Class B, characterized by predominantly uniform sequence patterns, an increase in filter depth had a more significant effect than the addition of multi-scale convolution via mCNN. In Class C (ABC), the multi-window CNN (mCNN) architecture demonstrated outstanding performance, achieving an MCC of 0.9331 and an AUC of 0.9948. This illustrates the efficacy of mCNN in capturing multi-scale sequence dependencies, which are crucial for modeling the functionally and structurally diverse characteristics of ABC transporters. The statistics collectively illustrate the robustness and adaptability of the GenEfflux architecture. The findings indicate that the integration of a multi-window convolutional approach with optimal filter depth improves model performance and generalizability across various efflux transporter families. As shown in Fig. 4, the multi-window CNN (mCNN) achieved superior classification performance across all efflux protein classes, with AUCs of 0.9893 for Class A, 0.9666 for Class B, and 0.9941 for Class C consistently outperforming both the single-window and fixed-filter CNN models. Supplementary Tables S1–S3 present comprehensive performance statistics for all architectural configurations, encompassing per-class evaluations of single-window CNNs, multi-window CNNs (mCNN), and models with differing convolutional filter depths. Figure S1 illustrates the AUC comparisons for the single-window CNN, while Figure S1 presents the AUC comparisons for the single-window CNN. Figure S2 shows the multi-window CNN comparisons, and Figure S3 illustrates the variations in filter depth. The findings highlight the significance of architectural choices, particularly the differences in kernel size and the modification of filter depth, in enhancing the classification performance of GenEfflux

Table 1
Performance of GenEfflux on 5-Fold Cross-Validation and Independent Test Sets Across Efflux Protein Classes.

Class	Evaluation Type	Sensitivity	Specificity	Accuracy	MCC	AUC
A	5-Fold CV	0.9151	0.9159	0.9156	0.8294	0.9623
	Independent Test	0.9672	0.9487	0.9568	0.9131	0.9737
B	5-Fold CV	0.9660	0.9640	0.9664	0.9330	0.9867
	Independent Test	0.9999	0.9231	0.9718	0.9327	0.9636
C	5-Fold CV	0.9288	0.9406	0.9346	0.8693	0.9773
	Independent Test	0.9667	0.9621	0.9642	0.9280	0.9812

Note: MCC: Matthews Correlation Coefficient; AUC: Area Under the ROC Curve.

Table 2
Performance of GenEfflux Under Different Architectural Configurations Across Efflux Protein Classes.

Class	Config Type	Config Value	Sensitivity	Specificity	Accuracy	MCC	AUC
A	Single Window	20	0.9262	0.9487	0.9388	0.8758	0.9839
	Filter	128	0.9672	0.9487	0.9568	0.9131	0.9888
	Multi Window	(Brown et al., 2020; Devlin et al., 2019; Yenduri et al., 2024; Wu et al., 2021; Ferruz, 2022; Hajiagha and Kafil, 2023)	0.9672	0.9423	0.9532	0.9061	0.9899
B	Single Window	20	0.9999	0.9038	0.9296	0.8459	0.9696
	Filter	64	0.9999	0.9615	0.9718	0.9327	0.9727
	Multi Window	(Brown et al., 2020; Devlin et al., 2019; Yenduri et al., 2024; Wu et al., 2021; Ferruz, 2022; Hajiagha and Kafil, 2023)	0.9999	0.9231	0.9437	0.8732	0.9696
C	Single Window	20	0.9667	0.9621	0.9642	0.928	0.9918
	Filter	64	0.9667	0.9621	0.9642	0.928	0.9924
	Multi Window	(Brown et al., 2020; Devlin et al., 2019; Yenduri et al., 2024; Wu et al., 2021; Ferruz, 2022; Hajiagha and Kafil, 2023)	0.9611	0.9716	0.9668	0.9331	0.9948

Note: MCC: Matthews Correlation Coefficient; AUC: Area Under the ROC Curve.

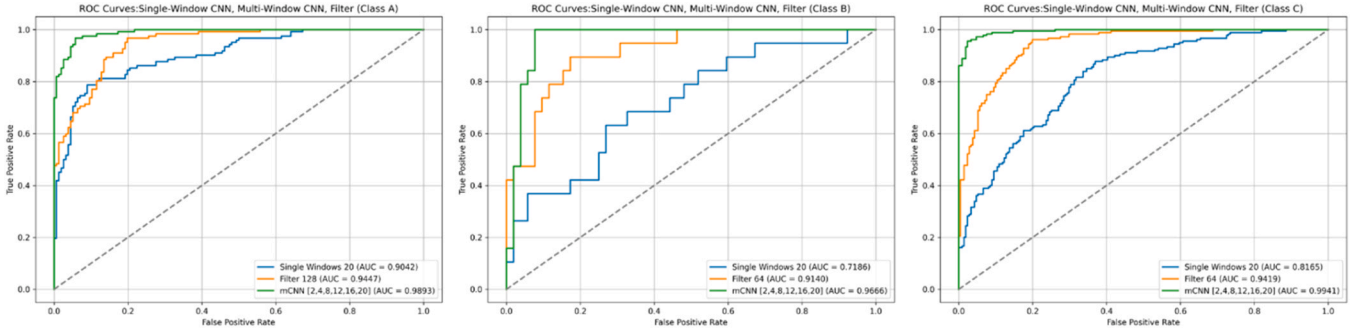


Fig. 4. ROC curves comparing the performance of single-window CNN, CNN with varying filter sizes, and the proposed multi-window CNN (mCNN) across efflux protein Classes A, B, and C. Each subplot represents one class.

for various efflux transporter types.

4.2. Performance comparison of generated Efflux Proteins with previous research

The performance of DeepEfflux and GenEfflux in predicting efflux proteins across three classes Class A, Class B, and Class C is presented in Table 3. The models are evaluated through sensitivity, specificity, accuracy, and Matthews Correlation Coefficient (MCC) to deliver a thorough analysis of their predictive performance. In Class A, both models demonstrate robust predictive performance, with GenEfflux exceeding DeepEfflux in sensitivity 0.9672 versus 0.9200 and MCC 0.9061 compared to 0.8883. DeepEfflux demonstrates marginally greater specificity 0.9683 compared to 0.9423, whereas GenEfflux attains similar accuracy 0.9532 versus 0.9545 with a more equitable predictive performance. Class B demonstrates a notable disparity, with deep efflux exhibiting a sensitivity of 0.5385, underscoring challenges in the identification of positive efflux proteins. The model demonstrates high specificity 0.94.48 and accuracy 0.9148; however, the Matthews correlation coefficient (MCC) is relatively low 0.4397. In contrast, GenEfflux exhibits complete sensitivity 0.9999 and notable specificity 0.9231, resulting in superior accuracy 0.9437 and MCC 0.8732. This suggests that GenEfflux provides a more thorough and equitable classification for Class B proteins. In Class C, GenEfflux demonstrates superior performance compared to DeepEfflux across all key metrics. The results demonstrate improved sensitivity 0.9611 vs. 0.9322, specificity 0.97.16 compared to 0.8276, accuracy 0.9668 relative to 0.8977, and a notably higher MCC 0.9331 compared to 0.7668. The findings demonstrate that GenEfflux offers a more reliable and accurate classification in this domain. To assess the effectiveness of the proposed GenEfflux model, we performed a detailed comparative evaluation against the baseline DeepEfflux model across three functional classes (A, B, and C). In Table 4 presents the averaged performance metrics obtained from five-fold cross-validation for both models, along p-values obtaining using paired *t*-test to assess statistical significance in prediction differences. The results indicate that DeepEfflux consistently outperforms GenEfflux in terms of raw performance metrics, particularly for Class B, where the difference is statistically significant ($p = 0.038$). For Class A, while DeepEfflux demonstrates slightly higher accuracy (0.9408 vs. 0.9199) and MCC (0.8817 vs. 0.8390), the performance difference is not statistically significant ($p = 0.091$). Interestingly, for Class C, GenEfflux achieves slightly better overall performance than DeepEfflux in several key metrics, including Accuracy (0.9434 vs. 0.9410), F1 Score (0.9452 vs. 0.9416), and AUC (0.9805 vs. 0.9789). Among these, only the difference in AUC was statistically significant ($p = 0.0076$), as determined by a paired *t*-test across the five validation folds. This suggests that GenEfflux offers competitive and stable performance in Class C and may even outperform DeepEfflux in certain evaluation dimensions.

Table 3
Performance Comparison of Generated Efflux Proteins Using deepEfflux and GenEfflux.

Class	Model	Sensitivity	Specificity	Accuracy	MCC
Class A	deepEfflux	0.9200	0.9683	0.9545	0.8883
	GenEfflux	**0.9672**	**0.9487**	**0.9568**	**0.9131**
Class B	deepEfflux	0.5385	0.9448	0.9148	0.4397
	GenEfflux	**0.9999**	**0.9615**	**0.9437**	**0.9327**
Class C	deepEfflux	0.9322	0.8276	0.8977	0.7668
	GenEfflux	**0.9611**	**0.9716**	**0.9668**	**0.9331**

Note: All values are reported in decimal format. MCC: Matthews Correlation Coefficient. Bold values in the final version should indicate best performance per class and metric.

Table 4
Comparison of GenEfflux and DeepEfflux across three functional classes with statistical significance (p-values).

Class	Model	Accuracy	F1 Score	MCC	AUC	Precision	p-value (vs DeepEfflux)
A	DeepEfflux	0.9408	0.9402	0.8817	0.9684	0.9232	–
	GenEfflux	0.9199	0.9268	0.8390	0.9653	0.9434	0.091
B	DeepEfflux	0.9920	0.9930	0.9823	0.9838	0.9881	–
	GenEfflux	0.9597	0.9633	0.9185	0.9849	0.9523	0.038
C	DeepEfflux	0.9252	0.8909	0.7846	0.9252	0.8888	–
	GenEfflux	0.9393	0.9411	0.8790	0.9783	0.9300	0.0076

*Statistically significant at $\alpha = 0.05$.

4.3. Amino acid composition across efflux protein superfamilies

To assess the biological plausibility of the generated sequences, we analyzed amino acid frequency distributions and PSSM similarities across all five major efflux protein families: MATE, RND, SMR, MFS, and ABC. The results demonstrated overall consistency between ProtGPT2-generated and natural sequences, particularly in hydrophobic residues such as alanine, leucine, and isoleucine, which are critical for stabilizing transmembrane domains. Importantly, several residues involved in conserved functional motifs were retained across families. For instance, glycine, known for its role in conferring conformational flexibility in SMR and MFS transporters, showed strong representation in generated sequences. Acidic residues like aspartate (D) and glutamate (E), which are essential for proton coupling in MATE and RND families, were present at comparable levels. Arginine (R), a key residue in the E/DR motif involved in proton-substrate symport in MFS and substrate-binding in ABC transporters, also appeared consistently. While minor deviations in frequency were observed for some residues Fig. 5, the generative model reproduced key biochemical and structural characteristics shared across natural efflux families. This suggests that ProtGPT2 captures not just statistical patterns but also functionally relevant sequence features across diverse transporter systems. To quantitatively assess these similarities, we calculated Jensen-Shannon divergence (JSD), cosine similarity, and performed chi-square tests for each family in Table 5. The MFS family yielded a JSD of 0.127 and cosine similarity of 0.967, indicating high correspondence. The RND and MATE families showed the greatest similarity, with JSD = 0.070 and cosine similarity = 0.989 for both. The SMR family presented a slightly higher divergence (JSD = 0.152, cosine = 0.955), though still within acceptable similarity bounds. The ABC family achieved a JSD of 0.077 and a cosine similarity of 0.984, reinforcing its strong alignment with the natural sequence distribution. All five families showed non-significant chi-square p-values ($p = 1.0$), confirming that no statistically significant differences exist between the amino acid compositions of natural and ProtGPT2-generated sequences. Together, these results demonstrate that ProtGPT2 generates efflux protein sequences that are not only syntactically plausible but also statistically and compositionally consistent with natural proteins across diverse transporter families.

4.4. UMAP-Based visualization of feature representations of GenEfflux

To assess the discriminative power of the learned features for efflux protein classification, we applied Uniform Manifold Approximation and Projection (UMAP) to the model-generated embeddings. The resulting two-dimensional projections for both training and test sets across three efflux protein subclasses (A, B, and C) are shown in Fig. 6, the top row illustrates UMAP projections for training data, while the bottom row shows the corresponding test set embeddings. Red and blue points denote samples from Class 0 and Class 1, respectively. For Classes A and C, the training set embeddings exhibit noticeable clustering and class-wise separation, suggesting the model successfully captures class-specific features. In contrast, Class B displays more overlap between the classes, implying that this subset may pose greater classification challenges. The test set projections generally follow the same distribution trends observed in the training data. While some reduction in class separation is evident especially in Class B the model still retains a reasonable degree of discriminative structure. These visual patterns confirm that the model learns biologically relevant representations that generalize across efflux protein subclasses.

4.5. Grad-CAM visualization of GenEfflux

To interpret the model's internal reasoning, Gradient-weighted Class Activation Mapping (Grad-CAM) was applied to visualize the regions of the input that contributed most to the classification of efflux proteins. Representative Grad-CAM heatmaps from test samples are shown in Fig. 7. Each heatmap illustrates the model's attention across the sequence and feature dimensions for a single protein sample. All samples belong to the efflux protein class (True = 1), and most were correctly predicted. Regions with warmer colors indicate higher relevance in the model's decision. The Grad-CAM results reveal that important sequence regions are predominantly located in the early to middle segments of the input. These regions likely capture sequence motifs that the model has learned to associate with efflux protein classification. While this suggests the model is focusing on biologically meaningful patterns, we cannot directly confirm that these correspond to known functional domains such as substrate-binding sites without additional validation. Further investigation such as aligning these regions with annotated domain databases (e.g., Pfam, InterPro), structural mapping, or experimental data like site-directed mutagenesis would be necessary to determine any functional correlation.

5. Limitations and future work

5.1. Lack of experimental validation

This study is based entirely on computational evaluation metrics such as AUC, MCC, and accuracy. No wet-lab experiments were conducted to confirm the biological validity of the ProtGPT2-generated sequences. As such, the structural and functional relevance of these sequences remains hypothetical. Future work should incorporate experimental validation through techniques such as binding assays, gene expression studies, or structure prediction using tools like AlphaFold to assess the real-world functionality of the generated proteins.

5.2. Imbalanced representation of efflux families

The dataset used for model training and evaluation is unevenly distributed across efflux protein families. For instance, ABC transporters are highly represented with 2394 sequences, whereas families such as SMR and MATE are underrepresented, with only 18 SMR sequences and 102 MATE sequences, respectively. This imbalance may bias the model toward better performance on the more abundant classes and limit its predictive accuracy for underrepresented families. Although class-weighting and computational rebalancing strategies were applied, future work should prioritize the development of more balanced datasets to ensure fair and robust model performance across all efflux protein subclasses.



Fig. 5. Relative amino acid frequency distributions for natural and ProtGPT2- generated sequences across the five efflux protein superfamilies: MATE, RND, SMR, MFS, and ABC.

Table 5
Statistical comparison of amino acid composition between ProtGPT2-generated and natural efflux protein sequences across five transporter families.

Efflux Family	Jensen-Shannon Divergence	Cosine Similarity	Chi-Square p-value
MFS	0.127	0.967	1.0
RND	0.070	0.989	1.0
MATE	0.070	0.989	1.0
SMR	0.152	0.955	1.0
ABC	0.077	0.984	1.0

5.3. Limited taxonomic diversity

The dataset, primarily sourced from the Transporter Classification Database (TCDB), may disproportionately represent well-characterized bacterial species or model organisms. This potentially limits the model's generalizability to efflux systems in underrepresented taxa, such as archaea or fungi. Future studies should expand taxonomic coverage to improve the robustness and applicability of the model across a broader range of biological systems.

6. Conclusion

This research introduces an innovative method for generating and assessing efflux proteins through the application of deep learning

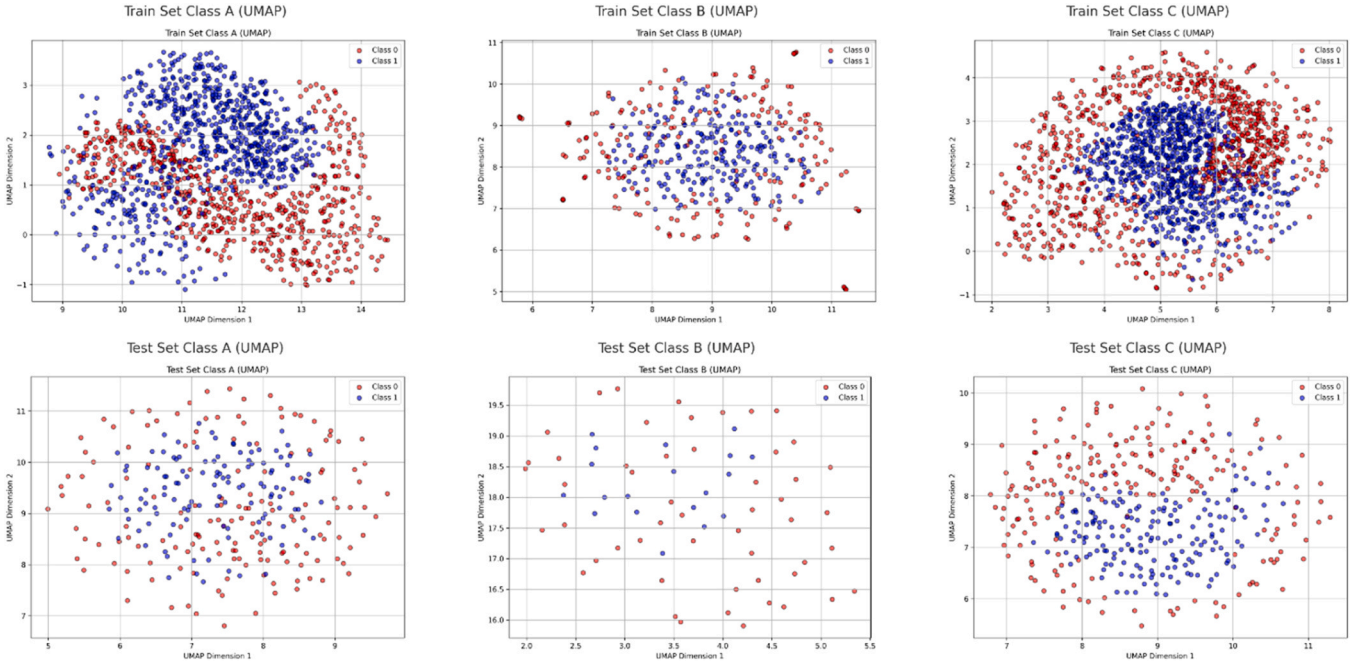


Fig. 6. UMAP projections of learned feature embeddings for GenEfflux. Top row: training data; Bottom row: test data. Red and blue points indicate Class 0 and Class 1 samples, respectively. Columns represent Class A, Class B, and Class C.

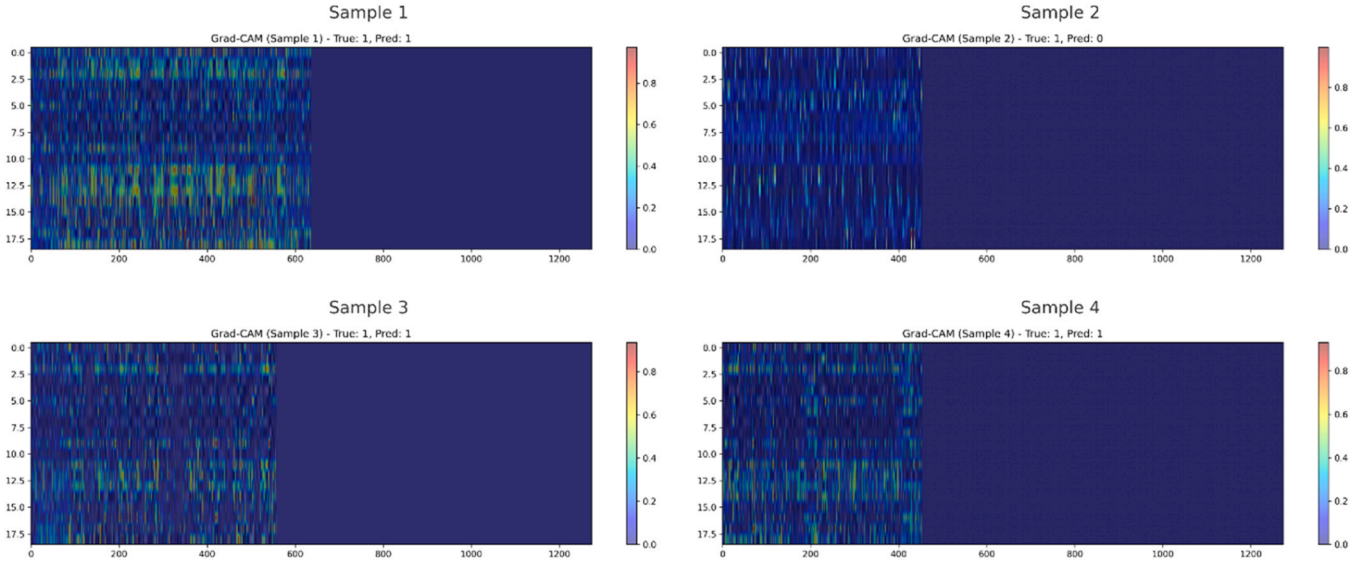


Fig. 7. Grad-CAM heatmaps for representative test set samples predicted as efflux proteins. Warmer colors indicate higher relevance of the sequence regions to the model's decision. The model exhibits consistent attention to early and mid-sequence motifs.

techniques. The method utilizes ProtGPT2 for sequence generation and a multiple-windows convolutional neural network (MCNN) for feature extraction from Position-Specific Scoring Matrices (PSSMs). The inclusion of generated sequences enhanced the dataset for all efflux families, leading to improved model performance. This generative method enables the examination of a broader spectrum of sequence variations, enhancing the assessment of how specific sequence characteristics influence efflux activity. The CNN-based feature extraction module of our method identifies complex patterns and relationships within generated sequences. Analyzing these sequences through a sophisticated CNN architecture can yield significant insights into the role of efflux proteins. Future research depends on the identification of potential regulatory motifs, interaction sites, and other functional elements essential for the proper functioning of efflux proteins; therefore, extracting this characteristic is critical. This research aims to improve the comprehension of protein regulation mechanisms in diverse biological systems. This study analyzes the interactions among various proteins, focusing on efflux proteins, their substrates, and other critical biological elements, to clarify the intricate regulatory mechanisms that control protein function. Understanding issues such as antibiotic resistance and protein misfolding and dysfunction is essential for addressing these challenges. The use of protein language models (PLMs) facilitates the examination of a wider range of protein families and their interactions, leading to the identification of new targets for drug development and therapeutic approaches. This study's findings may improve our comprehension of efflux protein dynamics and other proteins essential to biological processes. It is crucial to acknowledge that the findings obtained from this study are based solely on computational techniques. While sequences generated by ProtGPT2 show notable sequence-level similarity and classification effectiveness, additional validation like structure prediction through AlphaFold or experimental functional assays is necessary to confirm their structural integrity and biological activity.

Key Point

1) Introduces a novel deep learning framework that integrates ProtGPT2-generated protein sequences with multi-window convolutional neural networks (CNNs) to anticipate efflux protein functionalities.
2) Emphasizes the efficacy of pre-trained models such as ProtGPT2 to generate variable protein sequences that contain crucial biological characteristics.
3) Highlights the efficacy of multi-window CNNs in detecting essential amino acid motifs and evolutionary trends via the examination of multiple window sizes.
4) Offers an effective proof-of-concept for the integration of sequence generation with CNN-based feature extraction in predicting protein function.

CRediT authorship contribution statement

Quang Thai Ho: Writing – original draft, Visualization, Validation, Supervision, Methodology, Investigation, Data curation. **Yu-Yen Ou:** Validation, Supervision, Software, Methodology, Data curation. **Muhammad Hussain:** Writing – review & editing, Writing – original draft, Software, Methodology, Formal analysis, Data curation, Conceptualization.

Declaration of Competing Interest

The authors declare that there are no conflicts of interest related to this manuscript.

Muhammad Hussain declares no conflicts of interest in terms of financial relationships or affiliations that could influence the research.

Yu-Yen Ou declares no conflicts of interest in terms of financial relationships or affiliations that could influence the research.

Quang Thai Ho is the corresponding author and declares no conflicts of interest in terms of financial relationships or affiliations that could influence the research.

If any conflicts of interest arise after submission, they will be disclosed promptly.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.compbiolchem.2025.108595.

Data availability

Our code and dataset can be found on Github: https://github.com/mhb1717/efflux

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